File Tracker System

A PROJECT REPORT

Submitted by

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In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

in

Computer Engineering Department

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Gujarat Technological University, Ahmedabad

May 2023

Shankersinh Vaghela Bapu Institute of Technology

Vasan, Gandhinagar – 382650

CERTIFICATE

This is to certify that the project report submitted along with the project entitled **File Tracker System** has been carried out by **Patel Sahil Dineshkumar** under my guidance in partial fulfillment for the degree of Bachelor of Engineering in Computer Engineering Department, 8th Semester of Gujarat Technological University, Ahmedabad during the academic year 2022-23.

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Internship Certificate

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Date: 10 May 2023

CERTIFICATE

This Is To Certify That **Patel Sahil Dineshkumar** Has Successfully Completed His Project In **"File Tracker System"** Under Our Supervision And Guidance.

The Project Is Being Carried Out During The Period From 1-Feb-2023 To 09-May-2023 Towards The Partial Fulfillment Project In Website Development, Has Completed His Project Report.

He Has Completed The Project To Our Satisfaction.

We Wish His All The Best For Their Career.

From Tech infinity

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Vasan, Gandhinagar – 382650

DECLARATION

We hereby declare that the Project report submitted along with the Project entitled **File Tracker System** submitted in partial fulfilment for the degree of Bachelor of engineering in Computer Engineering to Gujarat Technological University, Ahmedabad, is a bona fide record of original work carried out by me at Tech Infinity, Gandhinagar under the supervision of Miss. Sejal Jadav and that no part of this report has been directly copied from any students' report or taken from any other source, without providing due reference.

Name of Student Sign of Student

Patel Sahil Dineshkumar

GUJARAT TECHNOLOGICAL UNIVERSITY

CERTIFICATE FOR COMPLETION OF ALL ACTIVITIES AT ONLINE PROJECT PORTAL

B.E. SEMESTER VIII, ACADEMIC YEAR 2022-2023

Date of certificate generation : 09 May 2023 (18:08:41)

This is to certify that, ***Patel Sahil Dineshkumar*** (Enrolment Number - 190750107031) working on project entitled with ***File Tracker System using Python with Django Framework and MYSQL backend*** from ***Computer Engineering*** department of ***SHANKERSINH VAGHELA BAPU INSTITUTE OF TECHNOLOGY, UNAVA, GANDHINAGAR*** had submitted following details at online project portal.

Internship Project Report	**Completed**

Name of Student : Patel Sahil Dineshkumar

Name of Guide : Miss.Mansuri Samina A.

Signature of Student : _____

*Signature of Guide : _____

Disclaimer :
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*Guide has to sign the certificate, Only if all above activities has been Completed.

ACKNOWLEDGEMENT

At the completion of my project, I would like to acknowledge all the persons, who make my project most successful.

My sincere thanks to my Professor **Miss. Samina Mansuri** for modeling my thoughts and vision towards all the subjects, I was studying in all these three years.

I take this opportunity to express my deep sense of gratitude and sincere thanks for the valuable guidance and motivation that I have received from my guide **Miss. Samina Mansuri,** professor, who has supported throughout my project completion. He made me aware about way to work in organization, which I always kept in my mind, and I am glad that results are positive for me.

I am very much grateful to **Miss. Sejal Jadav**, for accepting me to work under his training and supervision. He took prolonged interest in my work and directed me towards the predefined goal. He insisted for quality and expediency of software. I am pleased to work with such an indigenous. I thank him once again to approve my definition, which was infeasible at that time for me and also for keeping faith on me.

I am also greatly thankful to the entire **Computer Laboratory Staff** who have helped me in completion of this project directly or indirectly.

Your Sincerely,

Patel Sahil Dineshkumar

190750107031

ABSTRACT

File Tracker System is an application that manages all the file movement from one desk to another one and helps them in managing flow of files effectively and efficiently. All the files such as: reports, decisions, requests, reminders, and others can be processed and can also tracked by the system at any time. File Tracker System (FTS) is a web-based application which helps in tracking the movement of files. The application enables the users to maintain a consistent watch over the movement of various important documents in the process of decision-making. The system comprises of a centralized database that stores all the relevant information about the files, including their status and how many times this application is pending. A file tracking system provides timely and accurate information about the status of files, enabling users to track their files' progress.

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1. INTRODUCTION

1.1 OVERVIEW OF THE COMPANY

TECH infinity is providing IT solution, Training and Placement Services. TECH infinity involved in providing best quality talent to meet the needs of IT companies. It provides development service in different areas like Product Development (I.T Development),Skill Development (Training) and Career Development(Placements). It also create customize projects on the basis of client requirement and need. TECH infinity is the recruiting partner of more than 100+ companies.

It acts as a bridge between the companies and employees. It provides trained employees to the company which is more beneficial for the company in comparison to hire unskilled fresher. TECH infinity has created a talented Candidate who is dedicated to creating and supporting high quality products and services. TECH infinity is a global IT services company that creates, provides and supports software, Website and mobile app.

Our exhaustive portfolio of a host of services and solution enables TIS to match the customization and caters the needs of its clients for the challenges ahead.

TIS services, its team of people working for it and its relationship values cannot be felt by above few words. We invite you to call our office to experience in unleashing your technology challenges.

Our Values

- We believe in result-oriented work, keeping in mind the expectations of our clients.
- We work with positive approach with an intention to add value to our esteemed clients.
- We are honest, transparent and hold moral and ethical values above all.
- We believe, we are here to render dynamic, flexible, tailor made innovative solutions to provide immediate response and to take extra personal care of our clients. We offer precious service and expert solution to our clients, as we are not here in merely commercial business, but striving to build eternal professional relationship.

Our Vision

"To serve our clients by providing the highest quality of IT services that address their business issues. We attract, recruit, and retain the most knowledgeable and passionate professionals. We will continue to be known as the firm where personal attention will never become obsolete.

Our Mission

"To be the most trusted and respected IT firm recognized by our clients for delivering excellence."

1.2 WHAT IS WEB DEVELOPMENT

Web development is the process of creating websites or web applications. It involves a variety of tasks including designing, coding, testing, and maintaining web pages. Web development can range from creating a simple static website to developing complex web applications that require advanced programming skills. A web developer must have a strong understanding of web design, programming, and web standards to create effective and functional websites.

Web Development can be classified into three ways:

- Frontend Development
- Backend Development
- Full stack development

Frontend Development: Frontend development is a type of web development that focuses on the client-side or the user-facing part of a website or web application. Frontend developers use languages like HTML, CSS, and JavaScript to create the visual and interactive elements of a website. This includes designing the layout, styling the page, and implementing user interactions such as form submissions, animations, and responsive design.

Frontend Frameworks and Libraries:

- HTML
- CSS
- Bootstrap
- Javascript
- jQuery

Backend Development: Backend development is the process of building the server-side of a web application or website. It involves creating and maintaining the server, database, and application programming interfaces (APIs) that support the frontend user interface. Backend developers use programming languages like PHP, Python, and Ruby to handle data, user authentication, and other backend functionalities that allow a website or application to function correctly.

- **Python:** Python is a programming language that lets you work quickly and integrate systems more efficiently.
- **Django:** Django is open-source web framework that follows the model–template–views architectural pattern.

Full stack development: Full stack development is the practice of developing both the frontend and backend of a web application or website. Full stack developers have knowledge and expertise in a wide range of technologies, including HTML, CSS, JavaScript, server-side programming languages, databases, and server administration. They can handle all aspects of web development, from designing and developing the user interface to managing the server and database. Full stack development enables developers to create complex and highly functional web applications that deliver a seamless user experience.

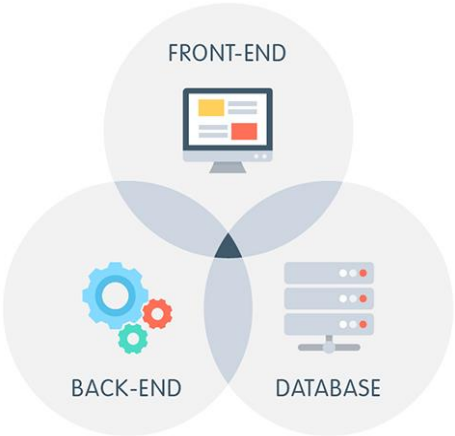

Figure: 1.1 Full stack Framework

1.3 DJANGO FRAMEWORK

Django is a high-level Python web framework that provides a robust set of tools and features for developing web applications quickly and efficiently. It follows the Model-View-Controller (MVC) architectural pattern and includes an Object-Relational Mapping (ORM) system for database management, making it a popular choice for building complex web applications.

Why Django Framework?

- Security: Django includes built-in security features, such as protection against common web attacks like SQL injection and cross-site scripting.
- Scalability: Django's modular design and scalability make it an excellent choice for building large and complex web applications.
- Rapid development: Django provides a range of pre-built components and tools, enabling developers to build web applications quickly and efficiently.
- Community: Django has a large and active community of developers, which means there are plenty of resources and support available for users.
- Versatility: Django can be used for a variety of web development projects, including content management systems, e-commerce websites, and social media platforms.

Django Architecture

Django architecture follows the Model-View-Controller (MVC) pattern, which is also known as the Model-View-Template (MVT) pattern in Django terminology. The architecture consists of three main components:

- **Model**: The model defines the structure and behavior of the data in the application. It interacts with the database and performs CRUD (Create, Read, Update, Delete) operations.
- **View**: The view handles the logic of the application and determines how data is presented to the user. It receives user requests, processes them, and returns the appropriate response.
- **Template:** The template defines the layout and presentation of the data, providing a user-friendly interface. It is responsible for rendering data and generating HTML pages that are sent to the user's browser.

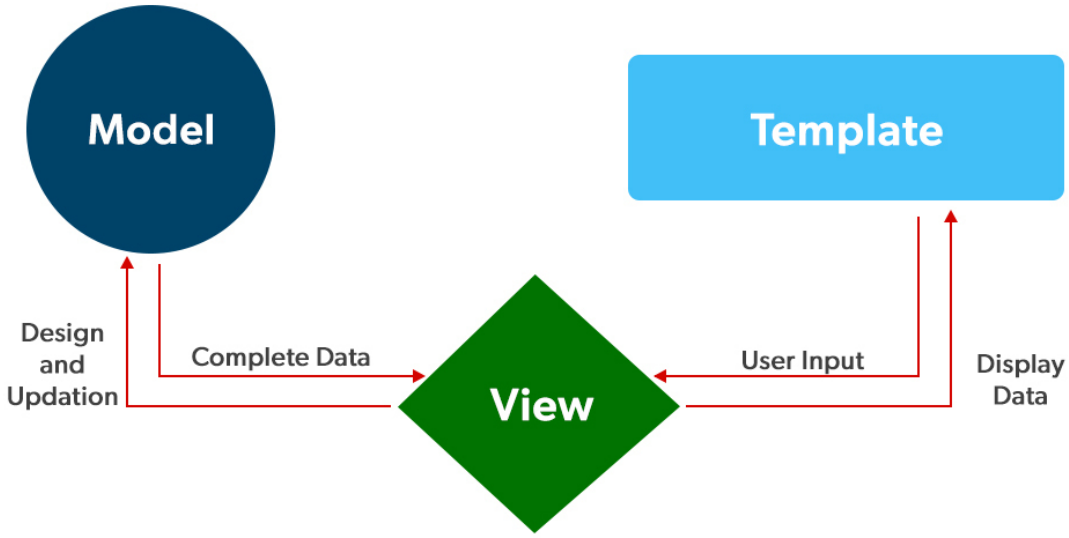

Figure: 1.2 Django Architecture

2. INTERNSHIP TRAINING PHASE

2.1 LEARNING FUNDAMENTALS

I began my training journey having an overview of the company. I went through the process of basic knowledge transfer at an early stage. I cleared my concepts in Python and Django framework.

Right from setting up the environment to learning the fundamentals of any programming language such as data-types, arrays, various looping constructs etc. I had hands on experience in object-oriented programming in Python, I implemented some Advanced concepts such as exception handling, OOPs concepts such as object-classes, abstraction, inheritance, polymorphism, encapsulation, multithreading concepts as well.

2.2 GETTING STARTED WITH WEB

I developed the understanding of how the web works under the hood, starting from how the world wide web got developed, how browser makes a query, how a DNS server works, how the resources are located over the internet, various protocols that are used to transfer variety of data. I then studied about how web pages are created using HTML and CSS.

I had hands on experience on HTML-CSS by making some static web pages, learned about some design fundamentals as to improve the web page design.

2.3 BUILDING BLOCKS OF WEB

2.3.1 HTML

I learned about various HTML semantics elements and how to use them, I learned about variety of html attributes, various form elements, meta tags and how they are useful in search engine optimization. I studied about good accessibility practices to implement while making websites so that it is easily accessible to all people. I also studied about search engine optimization and how to use meta tags to improve our website’s search ranking.

2.3.2 CSS

CSS can be used to design and improve the readability of our HTML elements. I learned different ways to associate the style sheet with the HTML file. I learned about CSS selectors, operators and their specificity. I understood about the box model, how to use inspect option in browser to manipulate the style of the component.

Meanwhile I also learned about ways to make our site responsive using various CSS attributes, CSS flexbox and CSS grid.

After having some strong knowledge of HTML and CSS, I look a leap towards the CSS framework called Bootstrap. Frameworks are used to abstract the complexity in out code and it makes out job easy.

2.3.3 JavaScript

I learned basics of JavaScript, on how to associate it with html file, I got familiar with its syntax, and learned how to declare variable, learned about execution context, scope chaining, arrays and its methods, various ways to loop, function, arrow function, how to create object and how to iterate over it, and also performed some programs for the same.

I also got myself familiar about DOM manipulation such as Accessing DOM Elements using "tag" name, Accessing DOM element using "classes", Accessing Elements using "id", Manipulating DOM Elements, Updating DOM styling with JavaScript.

2.3.4 Bootstrap

After I learned some of the concepts of JavaScript, I started to get familiar with one of the best free frontend open-source CSS framework i.e Bootstrap. It is the most popular HTML, CSS, and JavaScript framework for developing responsive, mobile-first websites.

I learned on how to link bootstrap to our HTML & CSS pages in multiple ways well it can be done in multiple ways like, through CDN links where it gets the data through the link from an external source of the bootstrap server, and one other simple way is to download the bootstrap to your local device and start using it via www.getbootstrap.com.

2.3.5 Python

Now it was time to get started on learning one of the top emerging programming language in the world. At first I started learning some basic concepts of python programming

language like - modules, exceptions, dynamic typing, very high level dynamic data types, and classes. Python is an interpreted, interactive, object-oriented programming language.

It was very fun to learn this programming language because it doesn't have any specific formatting for writing the code. You don't have to include and library to do mathematical calculations unlike in java which you have to include the libraries.

2.3.6 MYSQL

Whereas in other part you need to learn any one of the database to start making full stack website using Django. So I started to learn Mysql, there are many reasons why I choose Mysql, well you can use any database to configure the Django framework.

I learned the concepts of Mysql which includes – Data Types, Operators, Expressions, how to create/alter/drop DB. Some of the main operations in Mysql is CRUD operations i.e (*Create, Read, Update and Delete).* Conditions in SQL Database like – WHERE, AND, OR, LIKE, Limit and Order By, Group By, Having, Distinct and many other operations.

2.3.7 Django

I started to work in Django after 30 days of my internship journey, Django is a Python-based web framework that allows you to quickly create efficient web applications. It is also called batteries included framework because Django provides built-in features for everything including Django Admin Interface, database – Mysql, etc.

When you're building a website, you always need a similar set of components: a way to handle user authentication (signing up, signing in, signing out), a management panel for your website, forms, a way to upload files, etc. Django gives you ready-made components to use and that too for rapid development.

I learnt Django architecture in the right way with the help of my mentor first I started with MVT (Model-View-Template) architecture which is the pattern for developing a web application using Python as programming language and Django as framework. I learnt many topics such as – input/output operations, operators, data types, control flow, functions, data types, OOP's concepts, exception handling, file handling, Django templates, views in Django and basic app model.

3. SYSTEM ANALYSIS

After gaining good amount of knowledge about Python and Django web development technologies, I was introduced to a task of working on a site called “File Tracker System”.

3.1 INTRODUCTION

In today’s computerized world, administrative offices also tend to become paperless offices and with no more ‘piles of files’ scenario. This advancement aids the governing bodies in disposal of the issues within a reasonable time frame with evasion of stationery use. File Tracking System is a step in the direction of realizing the concept of paperless office.

The solution is a simulation of physical files into the electronic form. This enables fast retrieval and makes file movement & file tracking painless. Longevity of the electronic form solves the problem of material loss. It has a built-in dynamic workflow management system that capable of handling any organization’s hierarchy.

File Tracking System also captures and documents attached with the file and assignments for the file. It also allows users to put Notifications on the file. Complete history of the File Movement and Action Taken on the File is maintained.

3.2 OVERVIEW

File Tracker System is an application that manages all the file movement from one desk to another one and helps them in managing flow of files effectively and efficiently. All the files such as: reports, decisions, requests, reminders, and others can be processed and can also tracked by the system at any time.

File Tracking System (FTS) is a web-based application which helps in tracking the movement of files. The application enables the users to maintain a consistent watch over the movement of various important documents in the process of decision-making.

The system comprises of a centralized database that stores all the relevant information about the files, including their status and how many times this application is pending.

3.3 GOALS

A file tracker system should allow users to store all files in one location, making it easy to access and manage them. The system should provide automated workflows to help users move files through different stages of the document management process, such as approval, editing and In-process. File Tracker system should provide reporting and analytics capabilities to enable users to track document usage, monitor access.

3.4 PROBLEM DEFINITION

File Tracking System is an application that manages all the file movement from one desk to another one and helps them in managing flow of files effectively and efficiently. All the files such as: reports, decisions, requests, reminders, and others can be processed and can also tracked by the system at any time.

3.4.1 Existing System

- Today managing files in hard copy and searching them from bundles is a very tedious and burdensome task, causing tremendous confusion and pressure on the administrative body of the institutes.
- Movement of such files from one desk to another takes time and slows down the process.
- Also, it consumes a lot of space inside your cabin area and institutes need to provide extra facilities like storerooms to safeguard those files.

3.4.2 Proposed System

- The File Tracking System is an application that is designed for tracking the progress in each and every activity in the software development life cycle.
- The application can be available to the clients and the top management easily.
- The details like uploading, downloading, working days etc. can be tracked easily by the management without any difficulty.

3.5 FEASIBILITY STUDY

A feasibility study is a procedure for identifying, describing, and evaluating potential systems in order to choose the optimal one. The problem definition is solved during the study, and all components of the problem that will be included in the system are determined. The size of the project and the cost of benefits are also more precisely assessed. Before the project is planned or budgeted, a competent feasibility study will reveal the strengths and flaws in the system.

3.5.1 Technical Feasibility

It assesses whether the technology required for the proposed system is available, as well as how it can be integrated with the company's shareholder. The user's technical expertise must be assessed in order for the new technology to be understood and used.

Technical feasibility tries to answer the following questions to make the software feasible to developer: -

- The software or tools necessary for building or running the application are easily available or not?
- The compatibility amongst software exists or not?
- Are developers aware of these technologies?
- What about the alternative of these chosen technologies?

3.5.2 Economical Feasibility

The goal of the economic feasibility analysis is to establish the positive economic benefits that the proposed system will give to the company. Quantification and identification are included. It determines the financial rewards and costs connected with the system's development. The cost-benefit analysis is a term used to describe the economic feasibility of a project. It is vital to measure actual money value against activities required to deploy the system in order to conduct an economic feasibility analysis. We can assure that the expense of a potential new venture is ultimately advantageous for the people by using our system. As a result, we may conclude that it is financially feasible.

3.5.3 Operational Feasibility

Operational feasibility is a metric for how well a proposed system solves problems and exploits possibilities discovered during scope definition, as well as how well it meets the criteria identified during the requirement analysis phase of system development. It focuses on how well the proposed development project fits into the existing environment as well as the development schedule's goals. Human, organizational, and political factors all have a role in operational feasibility.

4. SYSTEM DOCUMENT

4.1 SCOPE

To develop a system using Python as programing language and Django as framework for the **File Tracker System** website. Frontend is developed using HTML, CSS, Java script and Bootstrap and Backend is managed using Python and Django Framework, database is developed using MySQL.

4.2 OVERVIEW

System design is the process of converting a logical representation of what the system must accomplish into a physical specification. The design creates a blueprint for the system and specifies how the various components interact. The design process follows a standard series of activities, starting with task evaluation and assignment and concluding with package design. The design phase is the part of the lifecycle where the detailed design of the system chosen in the study phase is completed.

4.2.1 Logical Design

Logic design refers to the component of the design process that is independent of any specific hardware or software platform. All functional characteristics of the system chosen for development in the analysis phase are documented during the logical design phase, regardless of the computer platform.

4.2.2 Physical Design

The logical specification of the system from logical design is transformed into technology-specific specifics from which all programming and system development may be completed. The system generates data output.

4.2.3 Database Design

The goal of database design is to offer auxiliary storage and contribute to the program's overall efficiency. One auxiliary storage medium must allow efficient data access. A database is an integrated collection of data that allows for centralized access to data from a programme. A database is a collection of logically connected data that is kept with the least

amount of redundancy possible in order to serve a large number of users quickly and efficiently.

4.2.4 Primary Key

A primary key is a unique identifier for all table records in a relational database table column (or set of columns). The following are the major characteristics of a primary key: For each row of data, it must have a unique value. It can't have any null values in it.

4.2.5 Foreign Key

A foreign key is a key that connects two tables. This is sometimes referred to as a reference key. A Foreign Key is a column or a set of columns whose values correspond to those of a Primary Key in another table.

4.3 E-R DIAGRAM

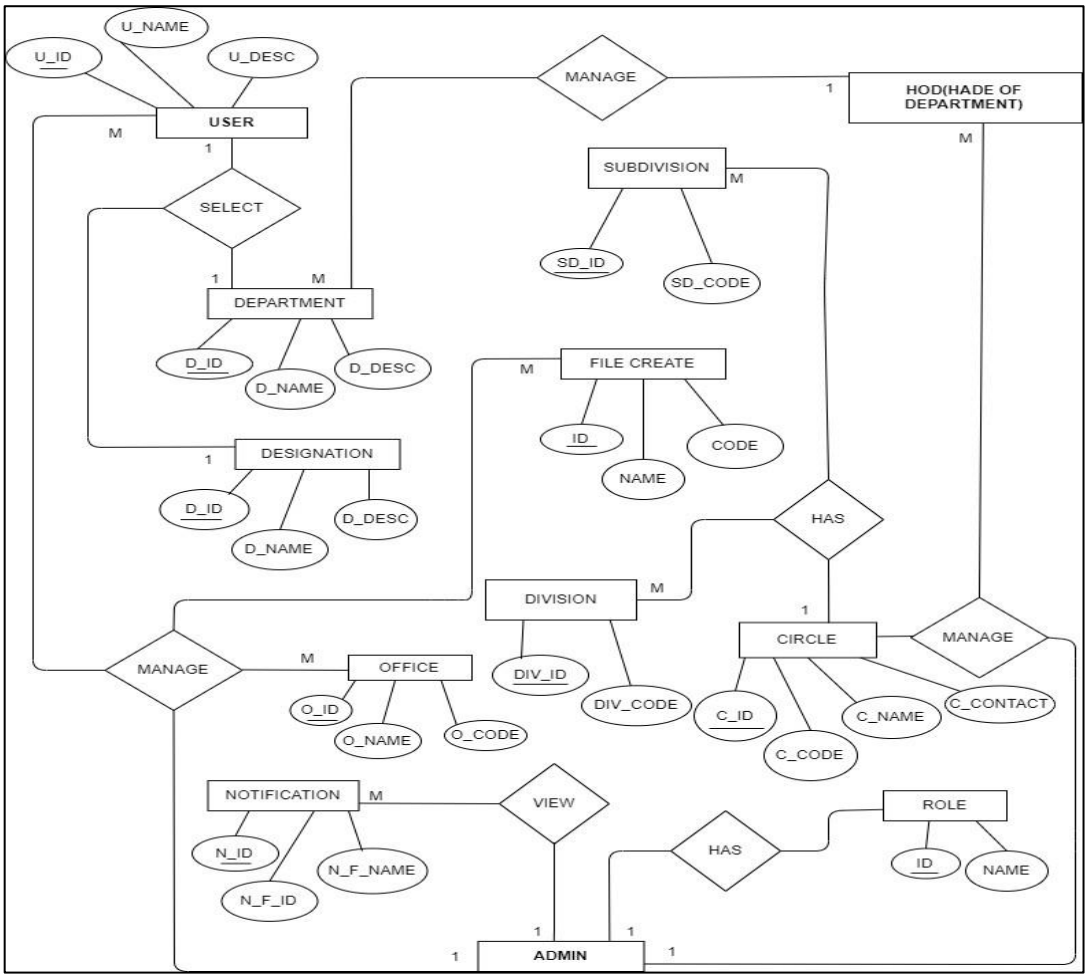

Figure 4.1 E-R Diagram

4.4 TABLES

Table 4.1 Circle

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
circle_code	int(11)	No	
circle_name	varchar(50)	No	
circle_contact	varchar(50)	No	
office_id	int(11)	No	

Table 4.2 Department

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
department_name	varchar(50)	No	
Scode	varchar(10)	No	
Cddept	varchar(1)	No	

Table 4.3 Design

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
Designation	varchar(30)	No	
Scode	varchar(10)	No	

Table 4.4 Division

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
division_code	int(11)	No	
circle_code	int(11)	No	

division_name	varchar(50)	No	
division_contact	varchar(50)	Yes	*NULL*
office_id	int(11)	No	

Table 4.5 File Create

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
Scode	varchar(10)	No	
Sid	int(11)	No	
dept_id	smallint(6)	No	
design_id	smallint(6)	No	
file_name	varchar(100)	No	
file_title	varchar(500)	No	
file_desc	Longtext	Yes	*NULL*
file_created_date	datetime(6)	No	
file_put_user_id	int(11)	No	
file_put_uname	varchar(50)	No	
file_status	varchar(10)	Yes	*NULL*
file_status_remarks	Longtext	Yes	*NULL*
file_status_date	datetime(6)	No	
last_update_user_id	int(11)	Yes	*NULL*
last_update_date	time(6)	No	
last_update_ip	varchar(20)	Yes	*NULL*
document	varchar(100)	Yes	*NULL*

Table 4.6 notification

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
file_id	int(11)	No	
file_name	varchar(100)	No	
sender_user_id	int(11)	Yes	*NULL*

sender_uname	varchar(50)	Yes	*NULL*
sender_date	datetime(6)	No	
receive_user_id	int(11)	Yes	*NULL*
receive_uname	varchar(50)	Yes	*NULL*
Action	varchar(20)	Yes	*NULL*
action_read_date	datetime(6)	No	
action_remarks	varchar(500)	Yes	*NULL*
action_date	datetime(6)	No	
last_update_user_id	int(11)	Yes	*NULL*
last_update_date	datetime(6)	No	
last_updated_ip	varchar(20)	Yes	*NULL*

Table 4.7 Office

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
office_code	int(11)	No	
office_name	varchar(50)	No	
office_contact	varchar(50)	No	
Uadserver	varchar(30)	No	
Udn	varchar(50)	No	
Uldaprdn	varchar(20)	No	
display_entry	int(11)	No	
short_code	varchar(3)	No	
cr_short_code	varchar(7)	No	

Table 4.8 Role

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
role_name	varchar(15)	No	

Table 4.9 Subdivision

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
office_id	int(11)	No	
subdivision_code	int(11)	No	
subdivision_name	varchar(50)	No	
circle_code	int(11)	No	
division_code	int(11)	No	
subdivision_contact	varchar(50)	No	

Table 4.10 User

Column	Type	Null	Default
id *(Primary)*	bigint(20)	No	
subdivision_code	int(11)	No	
division_code	int(11)	No	
circle_code	int(11)	No	
office_code	int(11)	No	
department_id	int(11)	No	
design_id	int(11)	No	
role_id	int(11)	No	
Uname	varchar(50)	No	
Password	varchar(20)	No	
Adminname	varchar(50)	No	
emp_no	varchar(10)	Yes	*NULL*
Email	varchar(100)	Yes	*NULL*
Phone	varchar(10)	Yes	*NULL*
Address	Longtext	Yes	*NULL*
City	varchar(15)	Yes	*NULL*
State	varchar(15)	Yes	*NULL*

Active	varchar(1)	No	
deactive_remarks	varchar(100)	Yes	*NULL*
deactive_date	datetime(6)	Yes	*NULL*
last_update_user_id	varchar(50)	Yes	*NULL*
last_update_date	datetime(6)	No	
last_update_ip	varchar(20)	Yes	*NULL*
Cdeptid	smallint(6)	No	

5. IMPLEMENTATION AND FEATURES

5.1 IMPLAMENTATION SOFTWARES

- HTML5
- CSS3
- Bootstrap
- Python 3.11
- JavaScript
- IDE (Visual Studio Code)
- MYSQL (Database)

5.2 SECURITY FEATURE

A file Tracker System typically includes several security features to ensure the confidentiality, integrity, and availability of the tracked files. Here are some common security features found in such systems:

- **Cross site request forgery (CSRF) protection: -**

  CSRF attacks allow a malicious user to execute actions using the credentials of another user without that user's knowledge or consent. Django has built-in protection against most types of CSRF attacks, providing you have enabled and used it where appropriate. However, as with any mitigation technique, there are limitations.

- **SQL injection protection: -**

  SQL injection is a type of attack where a malicious user is able to execute arbitrary SQL code on a database. This can result in records being deleted or data leakage. Django's query sets are protected from SQL injection since their queries are constructed using query parameterization. A query's SQL code is defined separately from the query's parameters.

- **Authentication and Authorization**: -

  Users are required to authenticate themselves before accessing the system. This can involve usernames and passwords authentication. Authorization mechanisms ensure

that users only have access to the files they are permitted to view or modify based on their roles and privileges.

- **Secure Transmission:**

  When files are transmitted between the user's device and the system, secure communication protocols like HTTPS (HTTP over SSL/TLS) are used. These protocols encrypt the data during transmission, protecting it from interception or tampering.

5.3 CODING STANDARDS

The coding standard is a well-defined and uniform coding style. With the aid of coding standards, anyone can look at any code and figure out what's going on, and newcomers can easily catch up. The way variables are named, the code is put out, the comments are specified, and the job of functions is carried out are some coding standard ways of handling numerous things. The coding standards that we employed in the software are described in this section. The tag is used in the context of coding standards.

6. WEBSITE MODULES

File Tracker System is classified into different modules:

- **Frontend**
  - User Module
  - HOD Module
  - Admin Module
- **Backend**

6.1 FRONTEND

Frontend is classified into three modules: Admin module, User module and Admin module.

6.1.1 User Module

User will go on visiting website in any browser and will register by clicking on register button. Then the user clicks on the login button and the user enters his username and password. If password is forgot then user reset the password.

Then the user will create a new file and forward it by writing the file title and information about the file. This file is go in one department into the other department to check. Then the user will check the status in notifications to see where his file has reached. User also change the email, phone and name.

6.1.1.1 Preview of the User Module

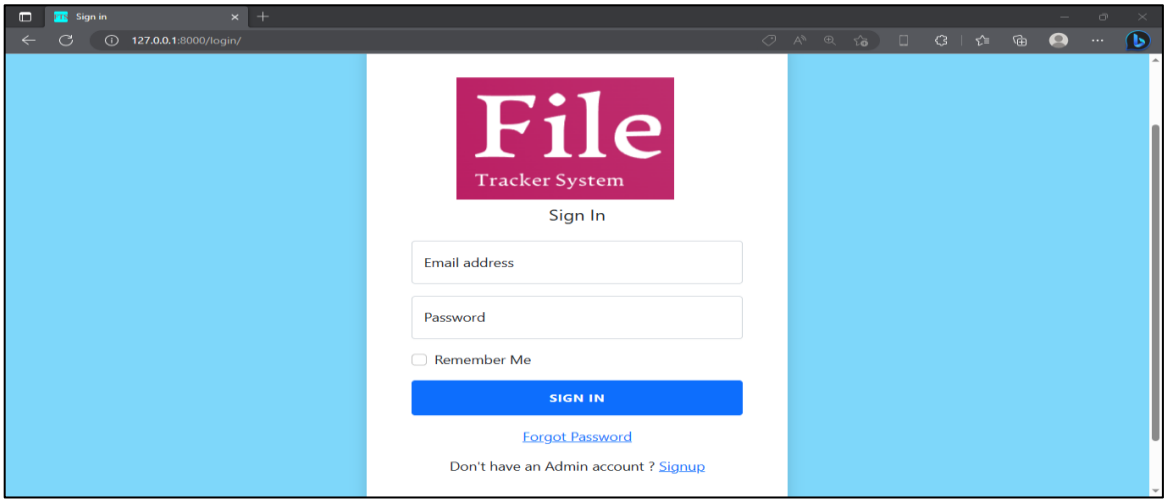

Figure 6.1 Login Page

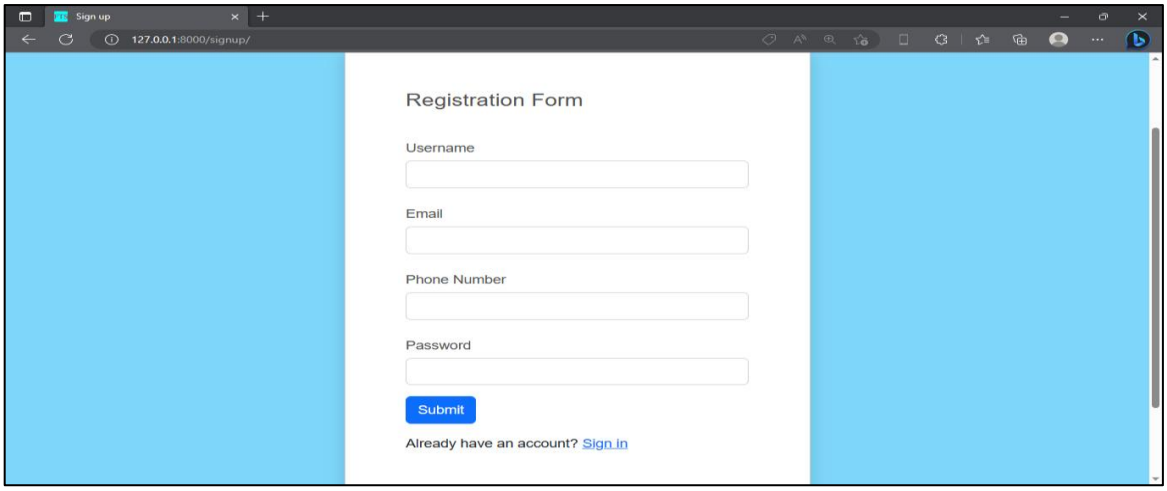

Figure 6.2 Signup Page

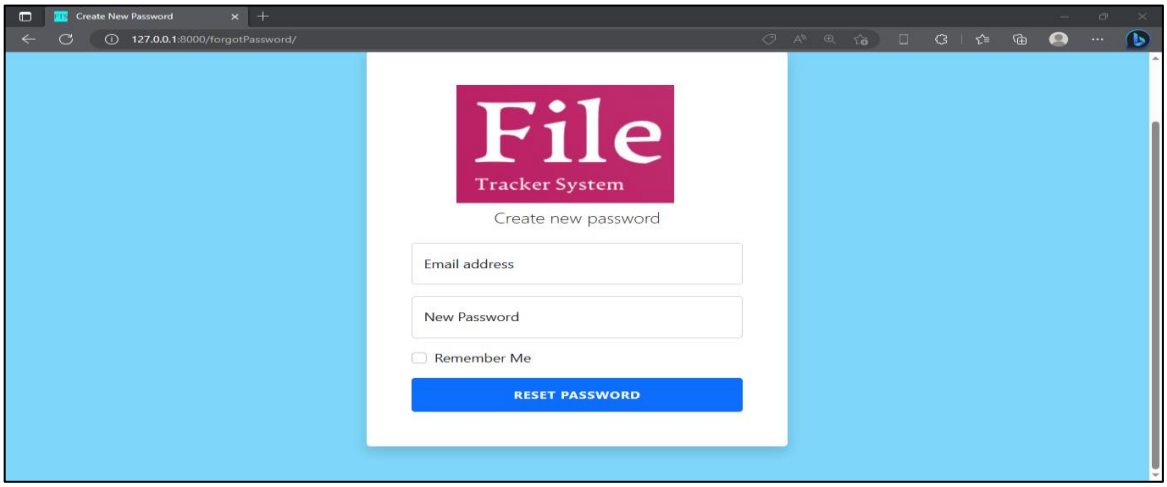

Figure 6.3 Forget Password Page

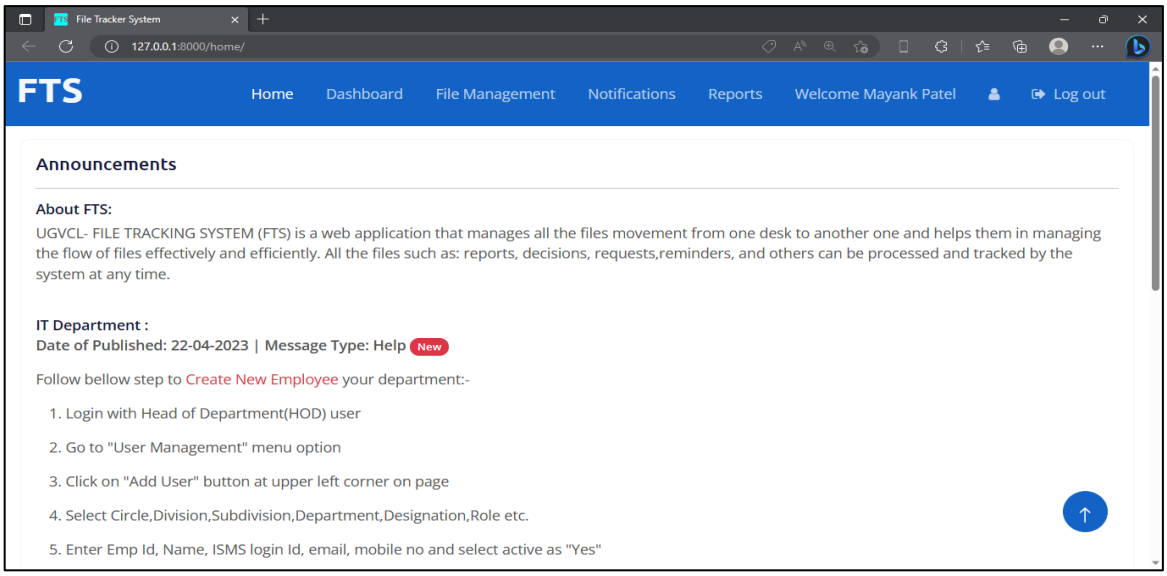

Figure 6.4 Home Page

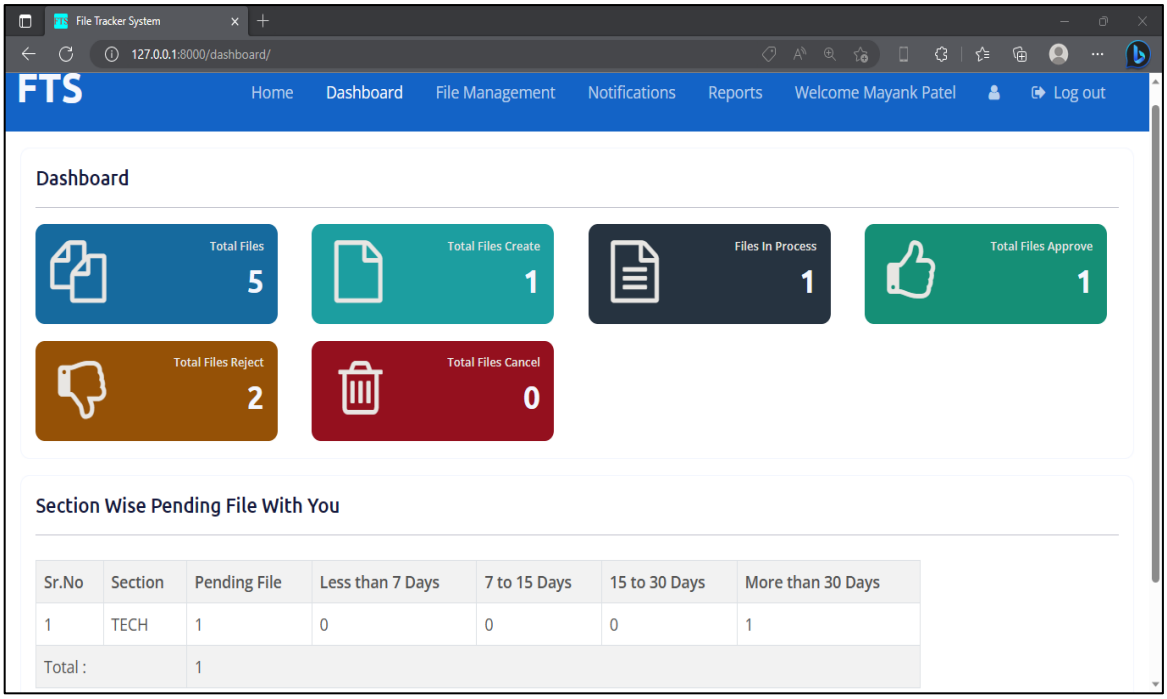

Figure 6.5 Dashboard Page

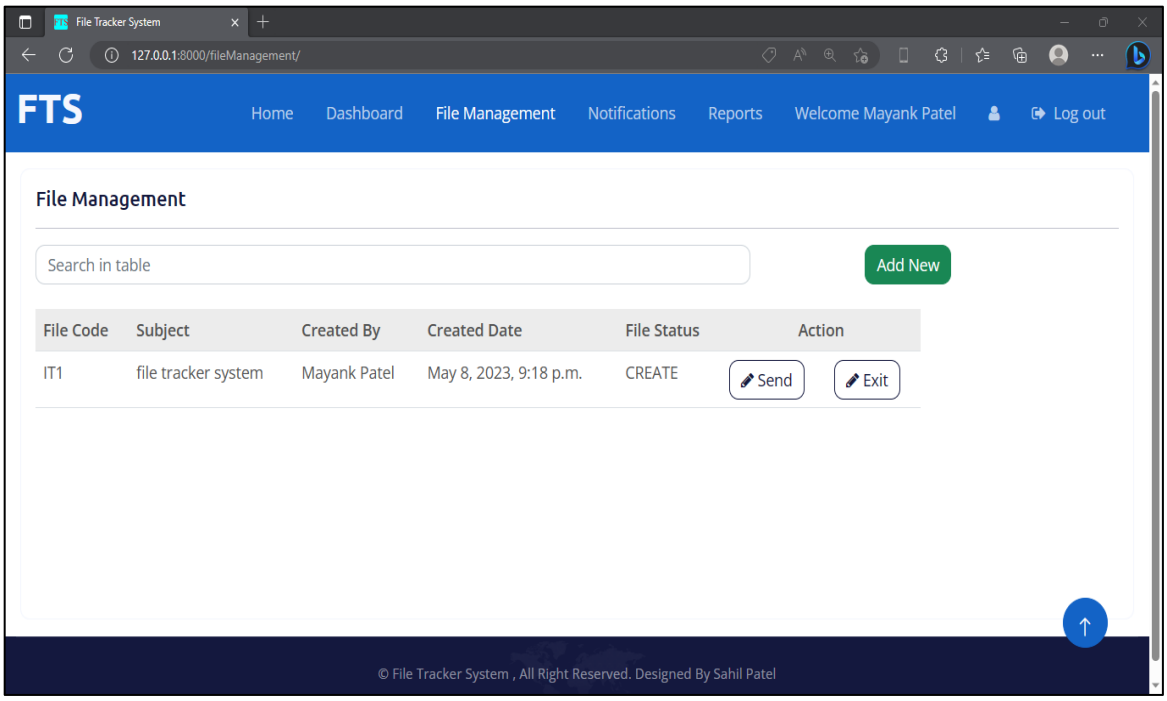

Figure 6.6 File Status Page

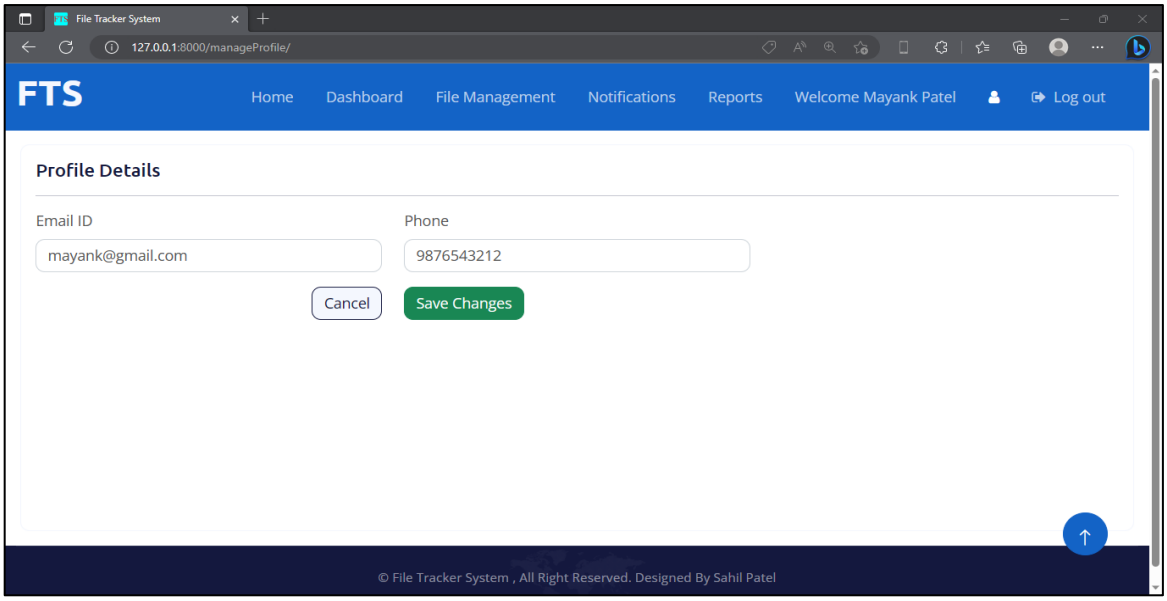

Figure 6.7 Profile Page

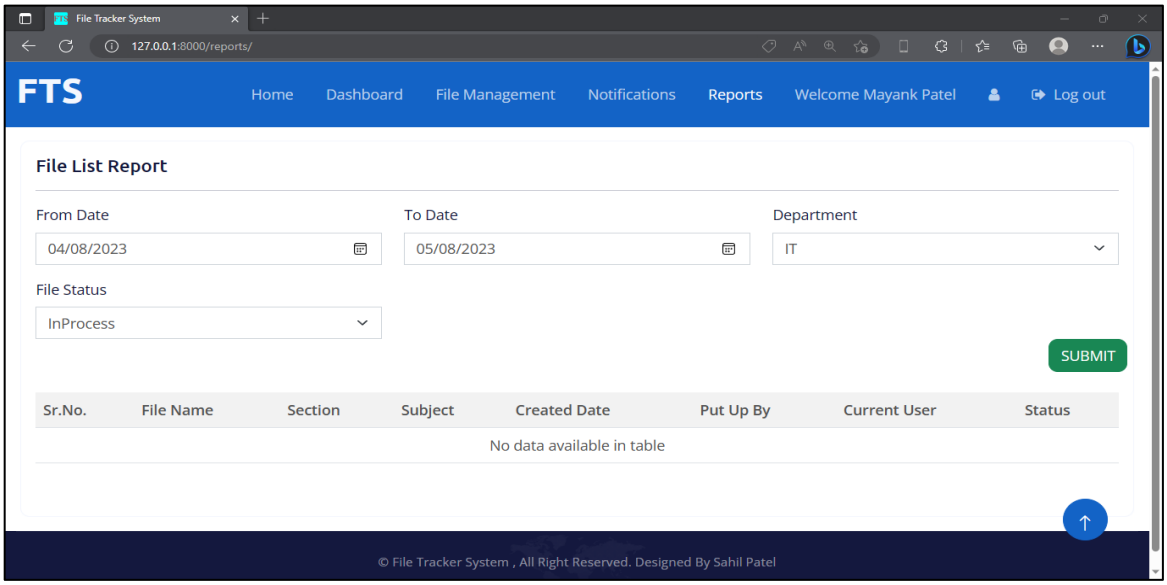

Figure 6.8 Report Page

6.1.2 HOD Module

This module typically provides the head of department with access to a dashboard where they can view all the files that are currently being processed by their department. They can also view the status of each file, such as whether it is pending, under review, or has been completed. The module may also allow the head of department to communicate with other departments involved in the file processing, such as forwarding files to other departments and receiving updates on the progress of those files.

6.1.2.1 Preview of the HOD Module

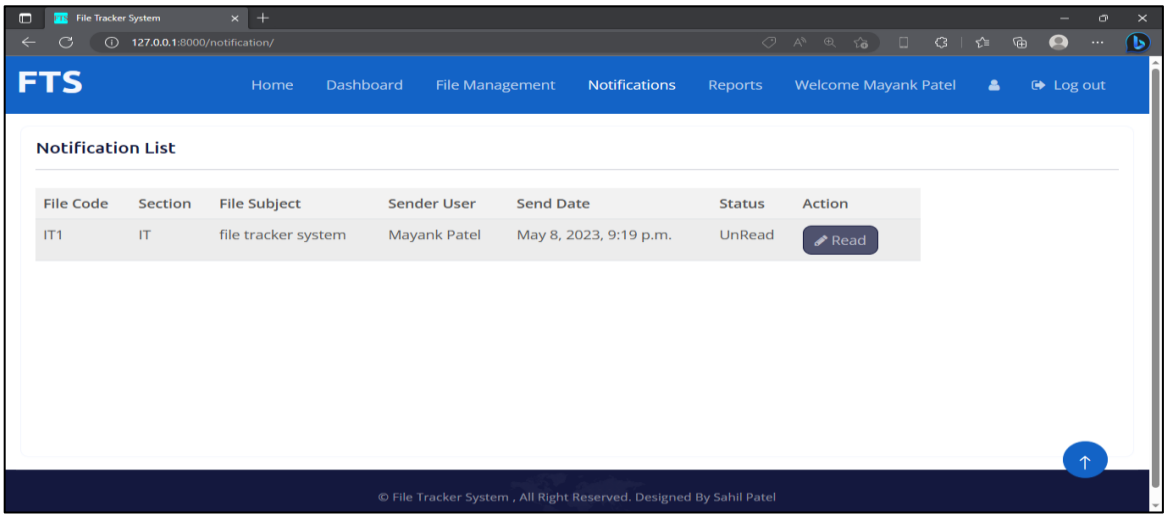

Figure 6.9 Notification Read Page

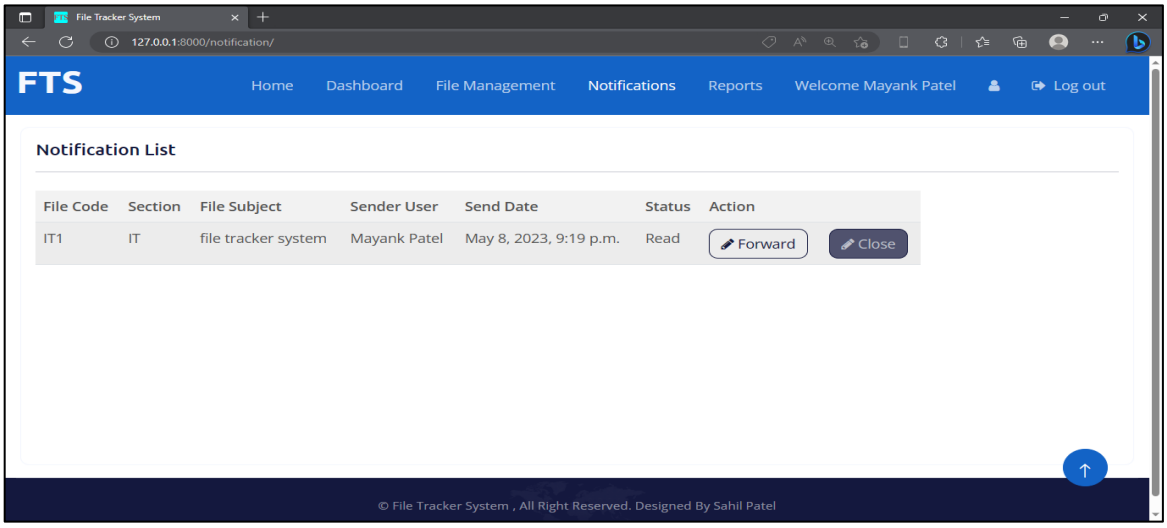

Figure 6.10 Notification Action Page

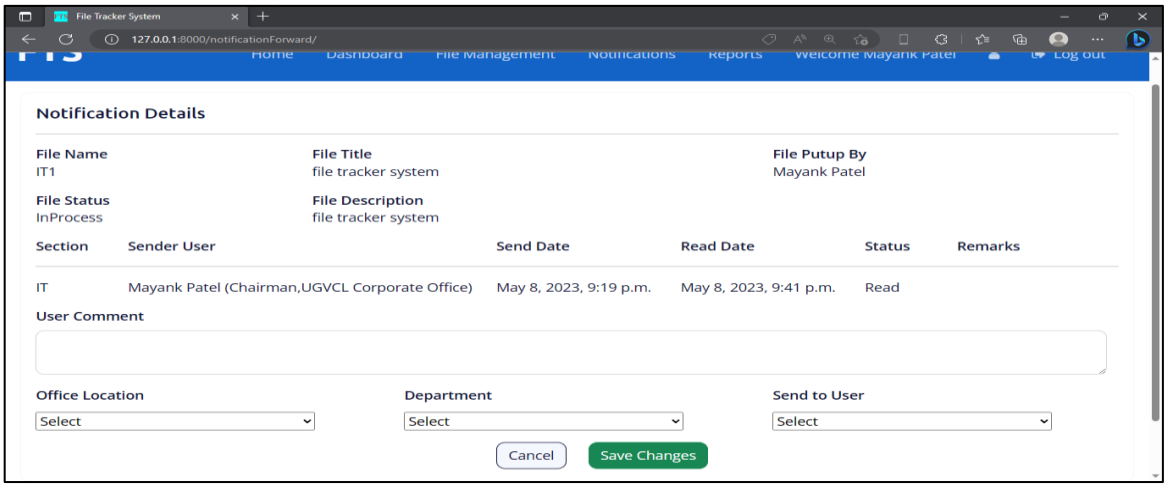

Figure 6.11 Notification Forward List

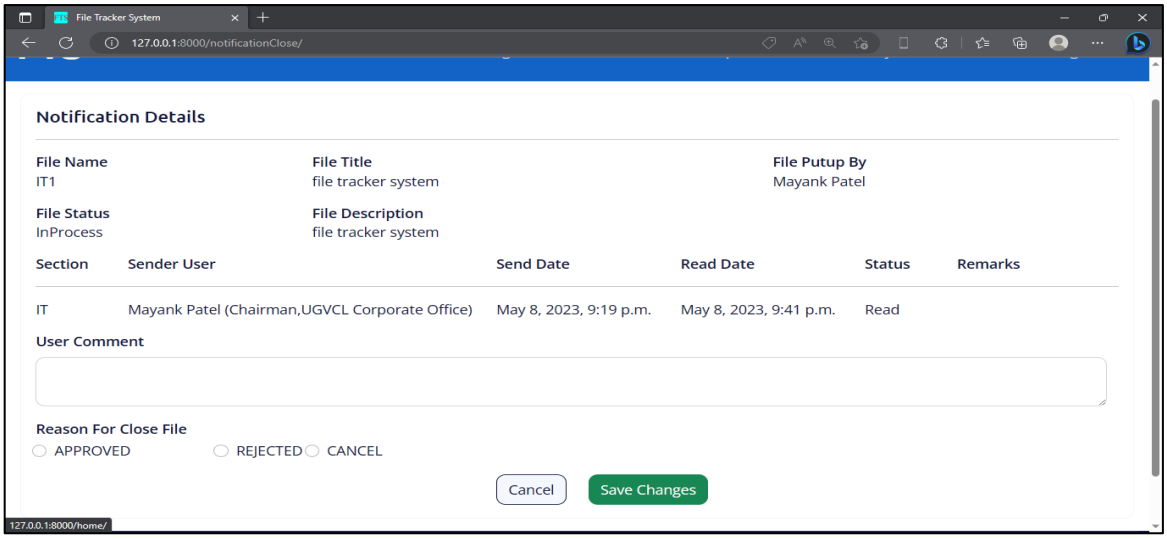

Figure 6.12 Notification Close Page

6.1.3 Admin Module

An admin module is an essential component of a file tracker system as it allows administrators to manage user accounts, set permissions, and oversee the overall operation of the system. Here are some of the key features and functionalities of an admin module in an online file tracker system:

- System configuration: Administrators can configure the system settings, including language preferences, notification settings, and other customizations.

6.1.3.1 Preview of Admin Module

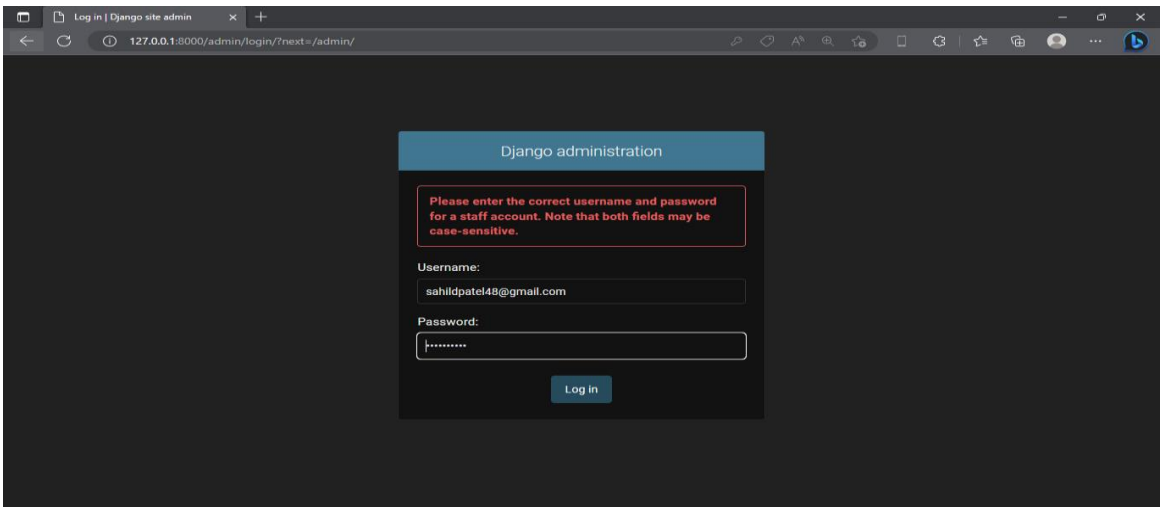

Figure 6.13 Admin Login

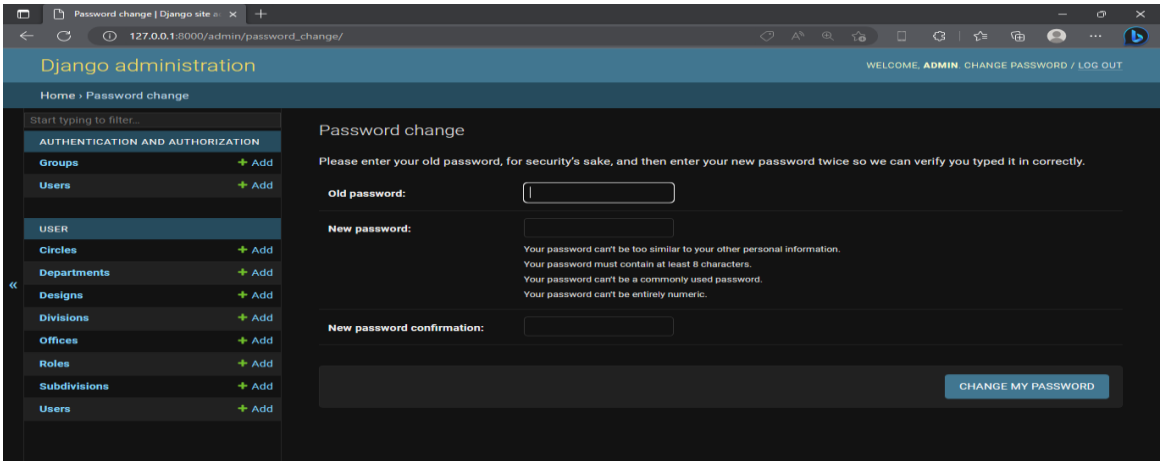

Figure 6.14 Admin Password Reset

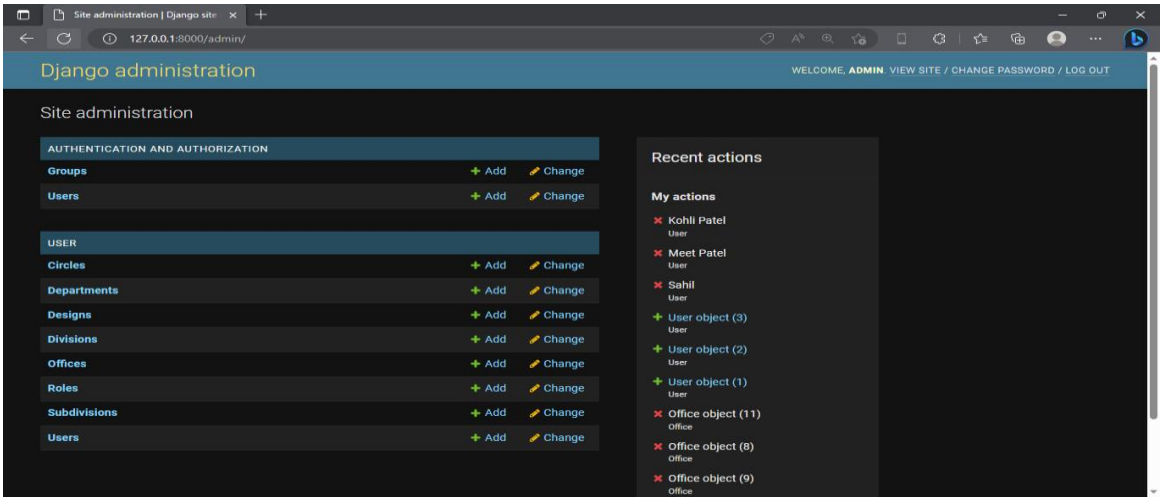

Figure 6.15 Admin Dashboard

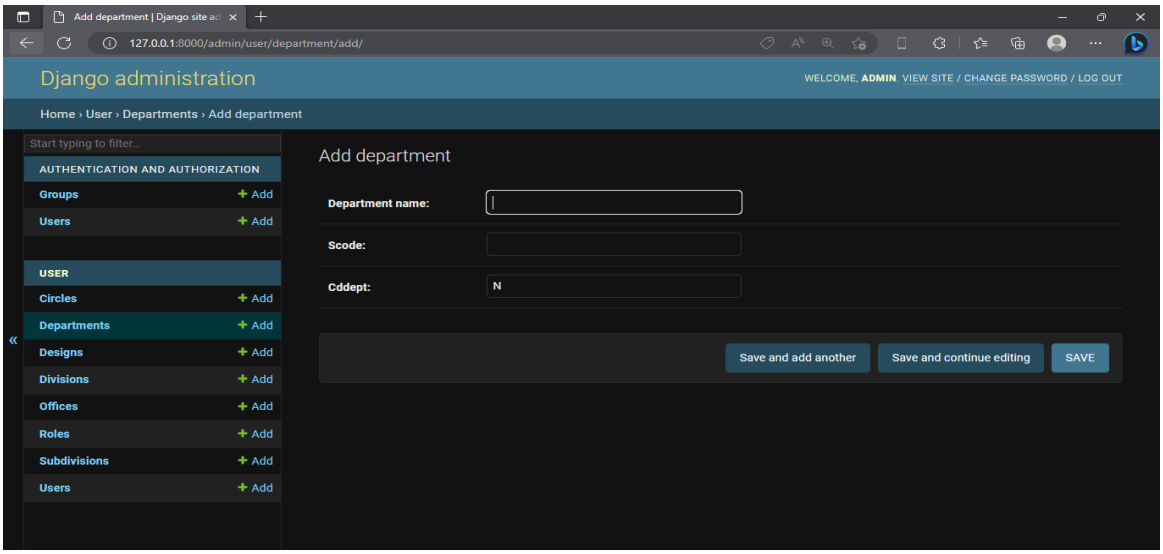

Figure 6.16 Admin Insert

6.2 BACKEND

The backend is developed using Python programming language, Django framework and SQLite as database this provides functionality of CRUD (Create, Read, Update & Delete) operations.

6.2.1 Code Snippets'

Here are the some of the snippets from the code which are a backbone to the website.

```
class File_Create(models.Model):
    scode = models.CharField(max_length=10)
    sid = models.IntegerField()
    dept_id = models.SmallIntegerField()
    design_id = models.SmallIntegerField()
    file_name = models.CharField(max_length=100)
    file_title = models.CharField(max_length=500)
    file_desc = models.TextField(null=True,default='null')
    file_created_date = models.DateTimeField()
    file_put_user_id = models.IntegerField()
    file_put_uname = models.CharField(max_length=50)
    file_status = models.CharField(max_length=10,default='create',null=True)
    file_status_remarks = models.TextField(null=True,default='null')
    file_status_date = models.DateTimeField()
    last_update_user_id = models.IntegerField(null=True,default='null')
    last_update_date = models.TimeField()
    last_update_ip = models.CharField(max_length=20,null=True,default='null')

    class Meta:
        db_table = 'File_Create'
```

Figure 6.17 Code Snippets File Create

```
class Notification(models.Model):
    file_id = models.IntegerField()
    file_name = models.CharField(max_length=100)
    sender_user_id = models.IntegerField(null=True,default='null')
    sender_uname = models.CharField(max_length=50,null=True,default='null')
    sender_date = models.DateTimeField()
    receive_user_id = models.IntegerField(null=True,default='null')
    receive_uname = models.CharField(max_length=50,null=True,default='null')
    action = models.CharField(max_length=20,null=True,default='null')
    action_read_date = models.DateTimeField()
    action_remarks = models.CharField(max_length=500,null=True,default='null')
    action_date = models.DateTimeField()
    last_update_user_id = models.IntegerField(null=True,default='null')
    last_update_date = models.DateTimeField()
    last_updated_ip = models.CharField(max_length=20,null=True,default='null')

    class Meta:
        db_table = 'Notification'
```

Figure 6.18 Code Snippets Notification

```python
class User(models.Model):
    subdivision_code = models.IntegerField(default='0')
    division_code = models.IntegerField(default='0')
    circle_code =  models.IntegerField(default='0')
    office_code = models.IntegerField(default='5')
    department_id = models.IntegerField(default='0')
    design_id = models.IntegerField(default='0')
    role_id = models.IntegerField(default='0')
    uname = models.CharField(max_length=50)
    password = models.CharField(max_length=20)
    adminname = models.CharField(max_length=50)
    emp_no = models.CharField(max_length=10,null=True,default="null")
    email = models.CharField(max_length=100,null=True,default="null")
    phone = models.CharField(max_length=10,null=True,default="0")
    address = models.TextField(null=True,default="null")
    city = models.CharField(max_length=15,null=True,default="null")
    state = models.CharField(max_length=15,null=True,default="null")
    active = models.CharField(max_length=1,default='Y')
    deactive_remarks = models.CharField(max_length=100,null=True,default='null')
    deactive_date = models.DateTimeField(null=True,default=0)
    last_update_user_id = models.CharField(max_length=50,default='null',null=True)
    last_update_date = models.DateTimeField()
    last_update_ip = models.CharField(max_length=20,null=True,default='null')
    cdeptid = models.SmallIntegerField(default='0')

    class Meta:
        db_table = 'User_Master'
```

Figure 6.19 Code Snippets User

6.3 DATABASE

This is the preview of database created using Mysql.

Table	Action	Rows	Type	Collation	Size	Overhead
admin_registration_master	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-
auth_group	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	32.0 KiB	-
auth_group_permissions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
auth_permission	Browse Structure Search Insert Empty Drop	88	InnoDB	utf8mb4_general_ci	32.0 KiB	-
auth_user	Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	32.0 KiB	-
auth_user_groups	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
auth_user_user_permissions	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
circle_master	Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
department_master	Browse Structure Search Insert Empty Drop	11	InnoDB	utf8mb4_general_ci	16.0 KiB	-
design_master	Browse Structure Search Insert Empty Drop	15	InnoDB	utf8mb4_general_ci	16.0 KiB	-
division_master	Browse Structure Search Insert Empty Drop	25	InnoDB	utf8mb4_general_ci	32.0 KiB	-
django_admin_log	Browse Structure Search Insert Empty Drop	113	InnoDB	utf8mb4_general_ci	48.0 KiB	-
django_content_type	Browse Structure Search Insert Empty Drop	21	InnoDB	utf8mb4_general_ci	48.0 KiB	-
django_migrations	Browse Structure Search Insert Empty Drop	57	InnoDB	utf8mb4_general_ci	16.0 KiB	-
django_session	Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	32.0 KiB	-
feedback_master	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	16.0 KiB	-

Console

Figure 6.20 MYSQL Database

7. TESTING

"Errors are more common, more pervasive and more troublesome in software than with other technologies"

Testing is the process of validating and verifying the artefacts and behavior of software under test. Product testing may also give an impartial, unbiased picture of the software, allowing businesses to realize and comprehend the risks associated with software deployment.

7.1 TESTING PLAN

A test plan serves as the foundation for a successful testing implementation. The testing plan outlines the general strategy for the test. The test plan, in many respects, acts as a summary of the test activities that will be carried out. It explains all of the tester's demands that must be addressed in order to effectively carry out the test and indicates how the tests will be structured. The purpose of test planning is to create a set of activities that, if completed, will identify all of the criteria in the programmer that have not been satisfied. There are several standards that may be utilized to create test programmer.

The testing effort is identified early in the deployment planning process, and the approach that your team will use to run tests. It also specifies the gear, software, and tools needed for testing, as well as the features and functions to be tested. A fully-rounded test strategy contains a testing timetable as well as any risk factors that may compromise testing. As a result, we can state that Test Planning specifies the activities, dependencies, and effort needed to run the system test.

7.2 TESTING STRATEGIES'

The test strategy is a systematic statement of how to test a software product. As needed, a test plan is designed for each level of testing. The test team analyses the requirements, develops the test strategy, and collaborates with the project team to assess the plan. The test plan may contain test cases, conditions, and the test environment, as well as a list of associated tasks, pass/fail criteria, and risk assessment. The testing strategy's objective is to set the overarching context for the whole testing process. The procedure varies based on

the unique aspects of your solution. This is the most crucial phase of the testing process in many ways since all subsequent testing decisions will be made within the framework of the strategy.

7.3 TEST CASES

A test case is a document that contains a set of test data, preconditions, expected outcomes, and post conditions that have been produced for a specific test scenario in order to check compliance against a given requirement.

The Test Case serves as the starting point for the test execution, and after applying a set of input values, the application has a definitive consequence and exits the system at some end point, which is also known as the execution post condition.

Case 1: -

Table 7.1 Test Case 1

Project Name	File Tracker System
Test Case ID	TC1
Test Item	Login page
Environment	Windows 10

Table 7.2 Test Case 1 Result

Sr No	Test Case Description	Input	Expected Output	Solution
1	User Name, Password : Mandatory fields	Username and Password	If the user name and password is right then login successfully	Result as expected

Case 2: -

Table 7.3 Test case 2

Project Name	File Tracker System
Test Case ID	TC2
Test Item	User Registration Page
Environment	Windows 10

Table 7.4 Test Case 2 Result

Sr. No.	Test Case Description	Input	Expected Output	Solution
1	User name, email id, phone number : **Mandatory Fields**	Click on Create My Account Button	Appropriate error message on click of "Save" button.	Result as expected
2	Email-id not proper	Enter E-mail Address	Appropriate error message on click	Result as expected
3	Phone not proper	Enter the Phone	Appropriate error message on click	Result as expected

8. CONCLUSION AND FUTURE PLANS

8.1 CONCLUSION

A file tracker system is a useful tool that enables users to track the progress of their files or documents in real-time. With this system, users can easily monitor the status of their files, including when they were last accessed, who accessed them, and any changes made.

The main benefit of a file tracker system is that it provides transparency and accountability in document management. This is particularly important in businesses and organizations where multiple people may be working on the same documents, and where it's essential to keep track of who has access to what.

8.2 FUTURE WORK

As per the requirements of the clients this project will be upgraded, more and more functionalities can be added. Couple of functionalities will be added in this website like mentioned below:

- Frontend will me made colorful as per the user’s requirement.
- Website will me make user friend website

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6. http://www.getbootstrap.com/
7. http://www.getbootstrap.com/docs/5.1/getting-started/introduction/
8. https://ugvcl.info/UGFTSclone
9. https://chat.openai.com/